

SESSION 7 — Process Synchronization and PATH Variable

Question 1

Write a C program to create a child process and use `waitpid()` to make the parent wait for the child process.



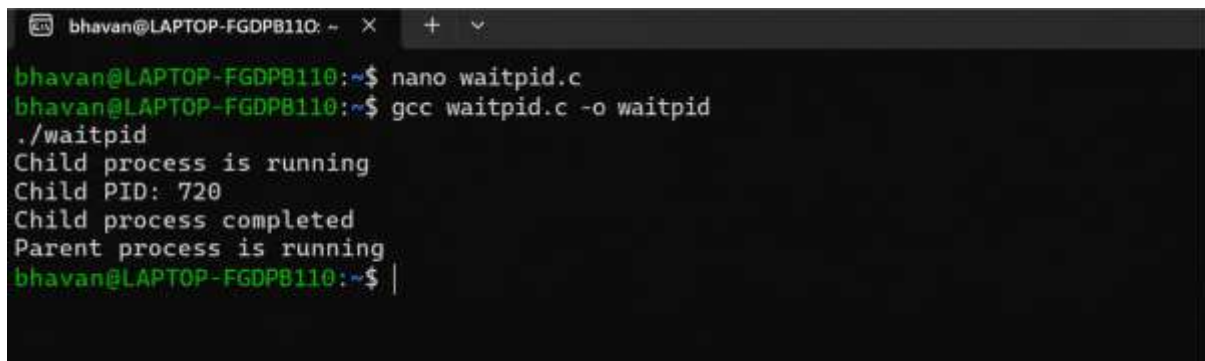
```
GNU nano 2.2 waitpid.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid == 0)
    {
        printf("Child process is running\n");
        printf("Child PID: %d\n", getpid());
    }
    else
    {
        waitpid(pid, 0);
        printf("Child process completed\n");
        printf("Parent process is running\n");
    }

    return 0;
}
```



```
bhavan@LAPTOP-FGDPB110: ~$ nano waitpid.c
bhavan@LAPTOP-FGDPB110:~$ gcc waitpid.c -o waitpid
./waitpid
Child process is running
Child PID: 720
Child process completed
Parent process is running
bhavan@LAPTOP-FGDPB110:~$
```

Question 2

Write Linux commands to display the `PATH` variable, find executable files, and check their permissions.

```

bhavan@LAPTOP-FQDPB118:~$ nano waitpid.c
bhavan@LAPTOP-FQDPB118:~$ gcc waitpid.c -o waitpid
./waitpid
Child process is running
Child PID: 728
Child process completed
Parent process is running
bhavan@LAPTOP-FQDPB118:~$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/ssl/lib:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Windows/System32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files/nod32:/mnt/c/Program Files/Git/cmd:/mnt/c/Users/bhavan/.local/bin:/mnt/c/Users/bhavan/AppData/Local/Microsoft/WindowsApps:/mnt/c/jdk-25_windows-x64_bin:/mnt/c/jdk-25:/mnt/c/jdk-25_windows-x64_bin:/mnt/c/jdk-25:/mnt/c/jdk-25_windows-x64_bin:/mnt/c/jdk-25:/mnt/c/Users/bhavan/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/bhavan/AppData/Roaming/npm:/mnt/c/Users/bhavan/AppData/Local/Microsoft/WindowsApps/Office/12.1.8.27438/clip.exe:/mnt/c/Users/bhavan/AppData/Local/Programs/AntiGravity IDE/bin:/snap/bin
bhavan@LAPTOP-FQDPB118:~$ which ls
/usr/bin/ls
bhavan@LAPTOP-FQDPB118:~$ which gcc
/usr/bin/gcc
bhavan@LAPTOP-FQDPB118:~$ type ls
ls is aliased to 'ls --color=auto'
bhavan@LAPTOP-FQDPB118:~$ ls -l /bin/ls
-rwxr-xr-x 1 root root 142112 Jan 23 2026 /bin/ls
bhavan@LAPTOP-FQDPB118:~$ command -v gcc
/usr/bin/gcc
bhavan@LAPTOP-FQDPB118:~$ |

```

waitpid() is used by a parent process to wait for a particular child process to finish. The PATH variable contains directories where Linux searches for executable commands. Commands like which and command -v help us find the location of commands.